

Muhammad Usman Javed

Graphics Programmer & Computer Science Student

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SUMMARY

Computer Science student focused on graphics programming and low-level systems, with hands-on experience building C++ desktop applications that combine real-time OpenGL rendering with algorithmic and data-processing tooling. Comfortable working from first principles, from direct framebuffer access in assembly up through modern shader-based rendering pipelines. Seeking a full-time remote internship in graphics, rendering, or C++ software engineering.

EDUCATION

B.S. Computer Science — Lahore Garrison University

Expected Graduation: 2028

Lahore, Pakistan · 5th Semester · CGPA 3.45

Relevant coursework: Data Structures & Algorithms, Computer Architecture, Linear Algebra, Theory of Automata, Design and Analysis of Algorithms, Probability & Statistics.

SELECTED PROJECTS

TwinCity3D — Real-Time 3D City / Digital Twin

- Built a real-time 3D city environment from scratch in C++/OpenGL, including buildings, road networks, a navigable camera, and atmospheric lighting laid out on a spatial grid to explore digital-twin style visualization.
- Implemented camera and view/projection systems and organized large scenes efficiently for real-time rendering.

C++, OpenGL, Real-Time Rendering

Theory of Automata Solver + NLU

- Developed a C++/ImGui application that parses natural-language automata problems and converts them into executable state-machine logic.
- Built a live, animated node/transition graph visualization that updates in real time as computation state changes.

C++, OpenGL, ImGui, NLU, Graph Visualization

Flappy Bird — 8086 Assembly (Low-Level Software Rendering)

- Wrote a fully playable game directly in 8086 Assembly using x86 real-mode DOS and VGA Mode 13h, writing pixels straight to the 0xA0000 VGA framebuffer.
- Implemented the full game loop, BIOS-interrupt input handling, and AABB collision detection with no engine or graphics API.

x86 Assembly, VGA Mode 13h, BIOS Interrupts, Software Rendering

Statistical Tool

- Built a modular C++ tool that parses datasets from files, computes statistical measures, and renders results as charts drawn entirely from scratch, with no external plotting library.

C++, File I/O, Data Structures, Algorithms

TECHNICAL SKILLS

Languages	C, C++, x86 Assembly, GLSL
Graphics	OpenGL 3.3, GLSL, ImGui, Render Pipeline, Shaders, Buffers (VAO/VBO/EBO), Textures
Foundations	Data Structures & Algorithms, Linear Algebra, Multivariable Calculus, Computer Architecture
Tools	Git, Visual Studio, Code::Blocks, MinGW

ACHIEVEMENTS

- Participated in a competitive programming event.
- Managed a vibe-coding evaluation event end-to-end — designed and wrote the assessment questions (scenario-based story puzzles testing vibe-coding ability) and led the evaluation process.